

4x4
SAR

Tech Preparedness Quick Reference

Apps, offline maps, battery, and satellite SOS.

NAVIGATION APPS (scan QR to download)

AllTrails+ (\$36/yr) Trail discovery + offline maps. Best for regular hikers. iOS Android

Gaia GPS (\$40/yr) Serious backcountry topo maps. Used by SAR teams nationwide. iOS Android

onX Backcountry (\$30/yr) Land ownership + hunting. Great offline topo maps. iOS Android

CalTopo (\$50/yr) Trip planning + custom maps. Best for route analysis. iOS Android

Organic Maps (FREE) Open-source offline maps. No account, no ads, no tracking. iOS Android

FREE OFFLINE MAPS (already on your phone)

- Google Maps: Search area > menu > Download offline map. Expires in 30 days.
- Apple Maps (iOS 17+): Profile > Offline Maps > Download New Map.
- Organic Maps: Open app, zoom to area, auto-downloads. Free forever.

BATTERY + POWER

- Low Power Mode: Enable at trailhead. iPhone: Settings > Battery. Android: Battery Saver.
- Screen brightness: Reduce to 30-40%. #1 battery drain on any phone.
- Close apps: Swipe away social media and unused background apps.
- Airplane mode: ONLY if you have a separate satellite communicator. Otherwise leave cellular ON.
- Power bank: Day: 5K mAh (1 charge). Overnight: 10K. Multi-day: 20K mAh.
- Cold weather: Keep phone + bank in inside pockets near body. Cold cuts capacity 20-40%.

SATELLITE EMERGENCY + COVERAGE

- Satellite 911 texting: FREE for all carriers via T-Mobile/Starlink (emergency 911 only). Full messaging: \$10/mo.
- iPhone 14+ SOS: Built-in satellite SOS, no subscription. Practice in Settings > Emergency SOS.
- Garmin inReach: Two-way satellite texts + SOS with 24/7 monitoring. ~\$15/mo. Best for backcountry.

Satellite 911 Signup (free) T-Mobile Coverage AT&T Coverage Verizon Coverage t-mobile.com/satellite-911 (QR/link opens full signup page) <https://www.t-mobile.com/coverage/coverage-map> <https://www.att.com/maps/wireless-coverage.html> <https://www.verizon.com/coverage-map>

Coverage maps are estimates. Actual signal varies with terrain, weather, and foliage. Most Cascade backcountry has NO cell service.



**Satellite 911
Signup**
[t-mobile.com/
satellite-911](https://t-mobile.com/satellite-911)



**T-Mobile
Coverage**
[t-mobile.com/
coverage/coverage-
map](https://t-mobile.com/coverage/coverage-map)



AT&T Coverage
[att.com/maps/
wireless-
coverage.html](https://att.com/maps/wireless-coverage.html)



**Verizon
Coverage**
[verizon.com/
coverage-map](https://verizon.com/coverage-map)